SHIKHA

ECE Engineer | Embedded Systems, Avionics, IoT & PCB Design

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PROFESSIONAL SUMMARY

ECE undergraduate at PSIT Kanpur specializing in embedded firmware (C/C++), avionics hardware, custom PCB design, and RF telemetry. Experienced in designing dual-MCU flight computers for sounding rockets, CanSat telemetry, and edge AI sensing systems. Winner of 2nd Prize at the IN-SPACe Model Rocketry Competition 2024-25.

EDUCATION

Pranveer Singh Institute of Technology (PSIT), Kanpur 2023 - Present

B.Tech in Electronics & Communication Engineering | Score: 78.6%

Kendriya Vidyalaya Sangathan 2022 - 2023

Class XII (CBSE): 77% | Class X (CBSE, 2020-21): 89.2%

ENGINEERING EXPERIENCE

Electrical Testing & Validation Intern - Raman Research & Innovation Pvt. Ltd. Mar 2025 - Jun 2025

- Tested and validated electrical circuits, sensor modules and embedded subsystems for system-level reliability.
- Assisted with PCB assembly, circuit validation, sensor interfacing and hardware debugging using bench oscilloscopes.
- Prepared comprehensive hardware validation reports and technical documentation for production builds.

Avionics & On-Board Computer Engineer - PSIT Vyomnauts (Team Sudarshan) 2024 - Present

- Designed and integrated dual-MCU flight computer (RPi Pico 2 + Arduino Nano 33 BLE) for rocket Trishul.
- Won 2nd Prize at the national IN-SPACe Model Rocketry Competition 2024-25 for avionics architecture.
- Core member of the IN-SPACe CanSat entry: won Best Design Award and Best Telemetry Data Award.

IoT & Robotics Workshop Trainer - College Technical Workshop (40+ Students) Jan 2026

- Conducted hands-on instruction in microcontroller programming, GPIO, ADC, PWM, and cloud IoT telemetry.

FEATURED PROJECTS & AWARDS

Rocket Avionics System (Trishul) [IN-SPACe 2nd Prize Award]

- Dual-controller architecture with Raspberry Pi Pico 2 primary MCU & Arduino Nano 33 BLE secondary.
- Integrated BNO085 IMU, BMP390 barometer, NavIC GPS, XBee S3B Pro RF telemetry and Quectel 4G backup.
- Custom 2-layer PCB designed in EasyEDA with MOSFET pyro triggers and isolated 7.4V to 5V/3.3V regulation.

MotorMind AI: Edge AI Motor Health Monitoring

- ESP32-based vibration (ADXL345), temperature (DS18B20), current (ACS712) and RPM sensor fusion node.
- On-device Decision Tree classifier executing anomaly detection with local OLED and remote Blynk alerts.

TECHNICAL SKILLS & TOOLCHAIN

Programming: Embedded C, C++, Python, Arduino C++

Microcontrollers: STM32, ESP32, Raspberry Pi Pico 2, Arduino Nano 33 BLE Sense Rev 2

Protocols & Comms: UART, SPI, I2C, FreeRTOS (basics), LoRa, XBee S3B Pro, 4G LTE, NavIC GPS

PCB & Hardware: EasyEDA, KiCad, Schematic Capture, DRC, BOM Preparation, Soldering, Multimeter

Software & Tools: STM32CubeIDE, MATLAB, Simulink, Multisim, XCTU, Git, GitHub